Subhadeep Ghorai

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Summary

SDE and System Design instructor building AI-agent evaluation systems and full-stack platforms on the MERN stack. Shipped a VS Code extension to the Marketplace using Node.js runtime internals, and a merged documentation fix to `electron/electron`. Open to SDE, New Grad and Instructor roles.

Experience

Newton School of Technology Dec 2025 – Present

SDE + Instructor 1

- Built AI-agent evaluation pipelines using Node.js and Claude API to autograde GitHub repos for 441 students in ∼110 minutes; evaluated 150+ student PRs and resolved 99 student grievances.
- Designed internal full-stack tools (React, Node.js, Express, MongoDB) for evaluation pipelines and operational workflows across the engineering program.
- Delivered 100+ technical lectures on Electron, React Native, System Design, OOP, and Software Engineering to 500+ engineers; 4.48/5 rating (91.55% CSAT).

Codingal Dec 2023 – Jan 2026

Teacher Mentor & Senior Coding Instructor *Remote*

- Designed and shipped **Atlas**, an operations platform (React, Node.js, Google Sheets API) replacing manual spreadsheet/Canva workflows for Codingal's teacher training programme; onboarded 100+ teachers in the first 10 days and automated 168+ certificates.
- Mentored 100+ students across 10+ countries through 2,500+ live sessions; promoted to Senior Coding Instructor (4.46/5 rating, 80% renewal rate).

Projects

EventLoop Studio | *TypeScript, Node.js vm, Acorn, React, VS Code Extension* [Marketplace]

- Published VS Code extension executing JavaScript in a real Node `vm` sandbox with native Promise intrinsics for spec-correct async behavior.
- Built scrubbable step-by-step execution replay engine capturing push/pop, console calls, timer/microtask/phase transitions as reconstructable execution steps.
- Modeled browser and Node.js (libuv phases) execution semantics for accurate cross-runtime visualization.

Application Development Course Management System | *React, Node.js, Express, MongoDB, Authentication*

- Built a full-stack course platform tracking 2,000+ projects across 500+ students on 2 campuses; teachers evaluate progress against a read-only, week-bucketed GitHub commit history.
- Designed JWT-authenticated, role-based views (student, teacher, admin) with isolated data; captured commits are immutable evaluation evidence.
- Implemented scheduled commit sync via GitHub Actions (instead of an in-process timer, since the free host sleeps), repository submission tracking, and a 485-problem statement bank.

Open Source

electron/electron – Merged documentation fix (PR #53114) documenting that pending navigation delays `debugger.sendCommand()` resolution (∼445ms) with timestamped logs.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java

Frontend Architecture: React, React Native, Electron; state management, component design patterns

Backend & Databases: Node.js, Express.js, MongoDB, PostgreSQL

Cloud & Infrastructure: AWS, Docker, Kubernetes, GitHub Actions, CI/CD pipelines

AI / LLM: Claude API, prompt engineering, tool use / function calling, MCP, RAG & vector search, LLM evaluation, agent orchestration

Education

VIT-AP University September, 2026

B.Tech in Electronics and Communication Engineering *CGPA: 8.52/10*